NNN-P-1475B
June 18, 1976
SUPERSEDING
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25 January 1968

FEDERAL SPECIFICATION

PAPER, FILTER, ANALYTICAL

This specification was approved by the Commissioner, Federal Supply Service , General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers analytical filter papers used in laboratories.

1.2 Classification.

1.2.1 Types and classes. Filter paper shall be of the following types and classes:

Type I - Qualitative
- Class 1 - For coarse and gelatinous precipitates
- Class 2 - For medium precipitates
- Class 3 - For fine precipitates

Type II - Quantitative
- Class 4 - For fine precipitates hardened
- Class 5 - For coarse and gelatinous precipitates
- Class 6 - For medium precipitates
- Class 7 - For fine precipitates
- Class 8 - For fine precipitates, thin.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

FSC 6640

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Federal Specifications:

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Box, Setup.

Federal Standards:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

Technical Association of the Pulp and Paper Industry (TAPPI) Standards:

T401 -- Fiber Analysis of Paper and Paperboard.
T413 -- Ash in Paper.
T429 -- Alpha, Beta and Gamma-Cellulose in Paper.
T430 -- Copper Number of Paper and Paperboard.
T435 -- Hydrogen Ion Concentrate (pH) of Paper Extracts.
T471 -- Testing of Analytical Filter Papers.

(Application for copies should be addressed to the Technical Association of the Pulp and Paper Industry, 360 Lexington Avenue, New York, NY 10017.)

3. REQUIREMENTS

3.1 Material. Material shall be of the highest quality throughout. Filter paper shall comply with requirements specified herein and shall be composed of 100 percent fully bleached fibers when tested as specified in 4.4.1.

3.2 Alpha cellulose. The alpha cellulose content shall be not less than 95 percent when tested as specified in 4.4.2.

3.3 Copper number. The copper number shall be not more than 0.5 when tested as specified in 4.4.3.

3.4 Acidity. The pH value shall be not less than 6.0 or more than 8.0 when tested as specified in 4.4.4.

3.5 Ash. The ash content of type I filter paper shall be not more than 0.15 percent. Type II filter paper shall be acid-washed for reduction of ash content. Ash content of type II, classes 4, 5, 6 and 7 shall be not more than 0.012 percent. Ash content of type II, class 8 filter paper shall be not more than 0.008 percent. Ash content shall be determined as specified in 4.4.5.

3.6 Bursting strength, filtration time and retention of precipitates. The bursting strength, filtration time and retention of precipitates shall comply with the requirements of table I, when tested as specified in 4.4.6, 4.4.7 and 4.4.8.

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TABLE I. Bursting strength, filtration time and retention of precipitates

Class	Bursting strength wet, minimum (points)	Filtration time (seconds)	Retention of precipitates (Filtrate clear of:)
1	3	20	Ferric hydroxide
2	3.5	40	Lead sulfate
3	4	150	Barium sulfate
4	60	260	Barium sulfate
5	3	20	Ferric hydroxide
6	3.5	40	Lead sulfate
7	4	150	Barium sulfate
8	4	150	Barium sulfate

3.7 Type II, class 4. Type II, class 4 filter paper shall have a surface sufficiently hard to permit precipitates to be scraped off.

3.8 Shape and dimensions. Shape of filter paper shall be circular or rectangular and shall have the specified dimensions. Diameter shall be specified for circular filter papers and side lengths shall be specified for rectangular filter papers (see 6.1). Tolerance on dimensions shall be $\pm$ 0.2 centimeters (cm.).

3.9 Count. Count shall be determined as specified in 4.4 9. No unit package shall contain less than 95 percent, and the average count of all packages shall be 100 percent of the specified count. Unless otherwise specified, unit package shall contain 100 filter papers.

3.10 Workmanship. Workmanship shall be first class throughout. The filter papers shall be free from defects listed in table IV and any other defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government,

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upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with table II and MIL-STD-105.

TABLE II. Sampling for examination

	Unit of product	Inspection level	AQL (percent defective)
For visual examination			
Major defects	Filter paper	II	1.0
dimensional examination	Filter paper	S-1	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with table III and MIL-STD-105. Acceptance number shall be zero, and inspection level shall be S-2.

TABLE III. Sampling for tests

Characteristic	Requirement	Test procedure
Fiber identification	3.1	4.4.1
Alpha cellulose content	3.2	4.4.2
Copper number	3.3	4.4.3
pH value	3.4	4.4.4
Ash content	3.5	4.4.5
Bursting strength	3.6	4.4.6
Filtration time	3.6	4.4.7
Precipitate retention	3.6	4.4.8
Count	3.9	4.4.9

4.3 Examination. Filter paper shall be examined to determine compliance with all requirements contained in this specification. Examination shall be in accordance with the defects listed in table IV but shall not be restricted to those defects.

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TABLE IV. Classification of defects

Category	Defect
Major:	
101	Filter paper not free of holes, tears, slits and unfinished edges.
102	Filter paper not free of integral foreign materials and discoloration.
103	Filter paper not free of surface dirt and grease.

4.3.1 Dimensional examination. Filter papers shall be examined for defects in dimensions. Any dimension not within the tolerance specified shall be considered a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Fiber identification. Fiber identification shall be determined in accordance with TAPPI T401.

4.4.2 Alpha cellulose. The alpha cellulose content shall be determined in accordance with TAPPI T429.

4.4.3 Copper number. The copper number shall be determined in accordance with TAPPI T430.

4.4.4 Acidity. The pH value shall be determined in accordance with TAPPI T435.

4.4.5 Ash content. The ash content shall be determined in accordance with TAPPI T413.

4.4.6 Wet bursting strength. The wet bursting strength shall be determined in accordance with TAPPI T471.

4.4.7 Filtration time. The filtration time shall be determined in accordance with TAPPI T471.

4.4.8 Retention of precipitates. Retention of precipitates shall be determined by examining the filtrate, which shall be caught in an Erlenmeyer flask, for any precipitate that may pass through the filter paper when 50 milliliters (ml.) of mixture is filtered. Swirl the filtrate to bring any precipitate to the center and view from above against a black background. The mixture shall be prepared as follows:

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4.4.8.1 Ferric hydroxide. Add a slight excess of dilute ammonium hydroxide to cold ferric chloride solution and shake. Make fresh each time and use immediately.

4.4.8.2 Lead sulfate, cold. Prepare a solution of lead acetate by dissolving 10 grams of the anhydrous salt or 11.5 grams (gm.) of the trihydrate in water and dilute to 100 ml. in a graduated cylinder. Filter through a class 7 filter paper and add 40 ml. of 6N sulfuric acid and 80 ml. of alcohol. Filter after standing for four hours. Wash the precipitate with dilute sulfuric acid (5 ml. concentrated acid in 100 ml. of water), then with alcohol.

4.4.8.3 Barium sulfate, hot. Dissolve 0.55 gm. of K_2SO_4 in 275 ml. of water and add 1.0 ml. of concentrated HCl. This solution contains the equivalent of about 0.74 gm. of $BaSO_4$. Heat to boiling, and while at the boiling temperature, add slowly and with continuous stirring 25 ml. of a 5 percent solution of $BaCl_2 \cdot 2H_2O$. It is very important to have the specified concentration of acid and to keep the solution at the boiling temperature during the precipitation. Let stand without agitation in a boiling water bath for two hours $\pm$ 15 minutes. Stir the precipitate until it is evenly suspended in the liquid.

4.4.9 Count. For packages where 100 is the specified count, the count shall be determined by weighing 10 of the filter papers and calculating the number per package, based on the weight of the total quantity in each package If the number of filters per package is not 100, 10 percent of the filter papers shall be weighed.

4.4.10 Examination of preparation for delivery. An examination shall be made to determine whether the packaging, packing and marking comply with the requirements of section 5. Defects shall be scored as specified in table V. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2, and the acceptable quality level shall be 4.0 percent defective.

TABLE V. Examination of preparation for delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence or method of application.

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4.10.1 Examination of closure, waterproofing and banding of shipping container. When shipping containers are required to comply with PPP-B-636, examination for defects in closure, waterproofing and banding shall be in accordance with the appendix of PPP-B-636.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B or C as specified (see 6.1).

5.1.1 Level A and B.

5.1.1.1 Unit package. The quantity of filter papers specified in 3.9 shall be packaged in a box of appropriate size and constructed in accordance with PPP-B-566 or PPP-B-676. Closure shall be adequate to prevent accidental opening.

5.1.1.2 Intermediate package. When specified, the quantity of filter papers of one description only shall be packaged in a box of appropriate size constructed in accordance with PPP-B-566, PPP-B-676, or PPP-B-636, class domestic.

5.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A. When specified, the quantity of intermediate packages shall be packed in a box of appropriate size and constructed in accordance with PPP-B-636, class weather-resistant, grade V3c. The closure, waterproofing and strapping shall be as specified in the appendix of the box specification.

5.2.2 Level B. The quantity of unit or intermediate packages of filter papers of one description shall be packed in a box of appropriate size and constructed in accordance with PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

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5.3 Marking.

5.3.1 Civil agencies. Marking for shipment shall be in accordance with Fed. Std. No. 123.

5.3.2 Military activities.

5.3.2.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement document:

a. Title, number and date of this specification.
b. National Stock Number (NSN).
c. Type and class required (see 1.2.1).
d. Shape and dimensions.
e. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
f. Quantity in unit package.
g. Quantity of unit packages in intermediate packages, and quantity of intermediate packages in exterior or shipping containers.

6.2 Classification change. Class 4 of NNN-P-1475A has been changed from type I to type II.

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